BULLETIN OF MARINE SCIENCE, 26(4): 417-432, 1976

AUTUMNAL REPRODUCTION IN *PANULIRUS ARGUS* AT BIMINI, BAHAMAS

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ABSTRACT

An extensive area near Bimini, Bahamas, was diver surveyed and monitored during five autumn periods to estimate and characterize the autumnal reproduction of *Panulirus argus*. Both the shallow Bahama Bank and the deeper fringing reef areas were sampled. Bank areas were totally devoid of any reproductive activity, while in local deep-water populations the reproductive rate far exceeded previously reported estimates; gravid females comprising 30-40% of the local female population. Correlations of female reproductive state with area of capture, depth of capture, and size of female are reported. Evidence is given that large females, which comprise a small segment of the population, are more fecund as a group than smaller, but still sexually mature, females. Data suggest that single fertilizations per spermatophore predominate in local populations in the fall.

Autumnal reproduction on *P. argus* may contribute more to the population than previously believed. It is felt that traditional trap capture has underestimated the gravid female population in certain habitats, and that diver-capture methods may be a necessary adjunct to stock assessment for this species.

Despite world-wide economic importance, little quantitative *in situ* work has been accomplished on the field ecology of the spiny lobsters (family Palinuridae). In particular, the great majority of work on the Florida spiny lobster, *Panulirus argus*, has been through examination of commercial catches, use of lobster traps, and other remote capture methods. Such research is subject to faulty interpretation since there exists the possibility of certain sampling errors. For example, individuals in all sexual states, molt conditions, and sizes may not be accurately represented in the sample because of differing propensities of trap entering (Morgan, 1974a, 1974b; Chittleborough, 1970; Heydorn, 1969c). More recently researchers have become aware of the importance of direct observation and capture in obtaining a less biased view of spiny lobster ecology. Haydorn (1969a, b, c), Berry (1971 a, b) and Silberbauer (1971a, b) used underwater observation and capture in the study of the African rock lobsters, and Olsen, Herrnkind, and Cooper (in press) investigated the behavior and population dynamics of *P. argus* at St. John, Virgin Islands, from the Tektite 2 manned undersea habitat.

A comprehensive, but qualitative, study on *P. argus* ecology in the Florida Keys was reported by Crawford and De Smidt (1922). Dawson and Idyll (Florida; 1951), Sutcliffe (Bermuda; 1952, 1953, 1956, 1957), Buesa Mas (Cuba; 1965), Paiva and Filho (Brazil; 1968), Butler and Pease (Panama; 1965), Peacock (Antigua and Barbuda; 1974) and others have extensively investigated the regional ecology of *P. argus* primarily using remote capture methods. Although the Little and Great Bahama Banks support an immense lobster population, both juvenile and adult, and a substantial portion of Florida landings in the past decade originated in the Bahamas (value $6.5 million in 1972; Simmons, 1974), research in this area has been sparse.

Past studies indicated that very low rates of reproduction were typical for *P. argus* during the autumn (< 5% gravid, Buesa Mas, 1965; approximately 5% gravid, Smith, 1951). The present study indicates that reproductive rates vary greatly with habitat, ranging from 0% gravid on the

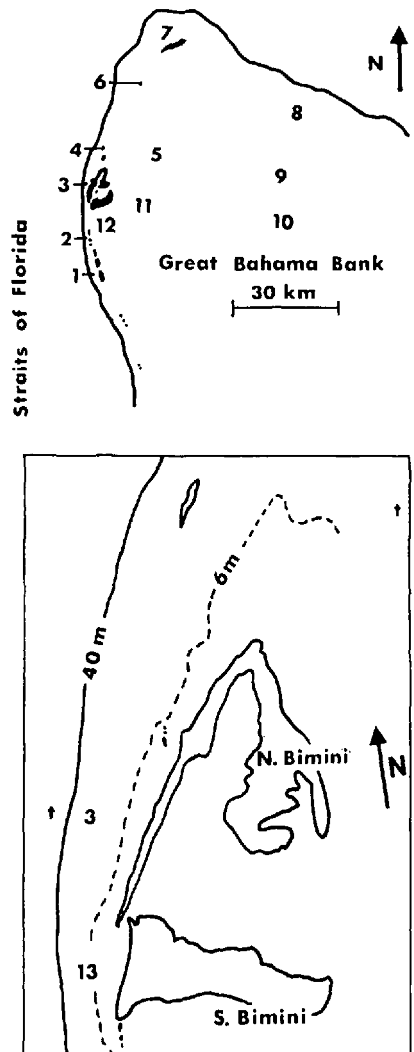


Figure 1. Chart of Bimini and surrounding area. Sample sites are: 1) Cat Cay, 2) Turtle Rocks, 3) Kinks, 4) Moselle Shoal, 5) "Cement wreck," 6) Hen & Chickens, 7) Great Isaac Light, 8) Gingerbread Ground, 9) Bank, 10) Mackie Shoal, 11) East Bank, 12) "Concrete Ship," 13) Bimini Harbor Patch Reef. Automatic thermal recording devices are indicated by a "t." Miami is approximately 90 km to the west.

Bahama Banks to 40% gravid on the deeper areas fringing the Bank during September.

Our own efforts included the use of SCUBA diving techniques in capturing and monitoring *P. argus*. The information presented here was obtained during efforts to characterize the autumnal mass migrations seen at Bimini, Bahamas (Herrnkind and Cummings, 1964; Herrnkind and McLean, 1971; Herrnkind et al., 1973) and was conducted in that area during autumns 1969, 1971, 1972, 1973, and 1974.

Sample Sites

Preliminary survey by underwater towboard and SCUBA diving on the extreme western edge of the Great Bahama Bank, north and south of Bimini (total length approximately 150 km, average depth 10-25 m), as well as the shallow Bahama Bank itself (2-10 m depth) was conducted to characterize the variety of habitats and the local Bimini spiny lobster population. Representative areas were chosen as typical of the habitat types found in the larger sample area and designated as sample sites; frequent diver visits, usually every 2 to 10 days, were made to observe and sample the resident and/or transient lobster populations (Fig. 1, Table 1). We attempted to capture all lobsters seen. Captives were examined and either tagged and returned or brought to the Lerner Marine Laboratory, Bimini, for further study. All available habitats were sampled, excluding the very shallow mangrove nursery areas adjacent to the islands.

The sampling site chosen as typical of the Great Bahama Bank was approximately 10 km east of South Bimini, 3-5 m in depth, about 2 ha in area (site 11). The substrate consisted of loose sand, while approximately 30% of the area was covered by sponges, sea whips, and sea fans where the lobsters obtained shelter. Few permanent rock or coral dens were available. Lobsters were numerous there until November, when the area became thinly populated. The

Table 1. Major sample sites

Area	Number in Figure 1	N	Collection dates	Substrate	Depth (m)	Shelter	Visibility (m)	Susceptibility to short-term temp., surge turbulence and turbidity fluctuations
Banks	11, 5, 10	235	Sept. 3–Nov. 25	Sea whips, fans, loose sand, wreckage at site 5	4–6	(marginal) bases of whips fans and sponges, wreckage at 5	2–10	High
Turtle Rocks	2	154	Aug. 29–Nov. 23	Sand, coral and rock outcrops	7–20	(very good) rock and coral dens and ledges	5–25	Medium
Bimini Harbor Patch Reef	13	746	Aug. 30–Nov. 19	Sand, low rock some coral	6–8	(good) low rock ledges	4–25	Medium High
Kinks	3	645	Sept. 3–Nov. 20	Large rock-coral outcrops	18–21	(very good) large rock-coral dens, ledges	10–30	Low

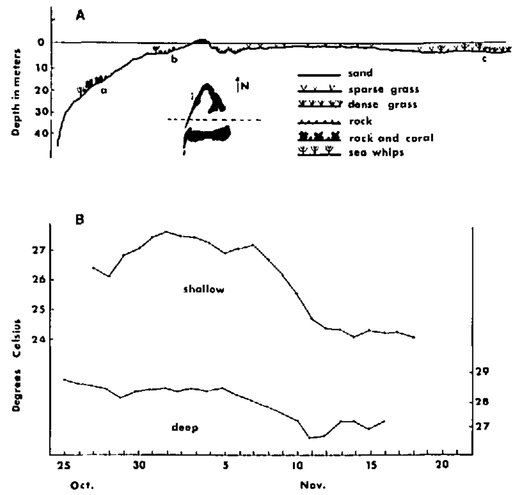


Figure 2. A. Cross section through Bimini showing depth characteristics and habitat types. a) Kinks area, b) Bimini Harbor Patch Reef, c) East Bank. To the west the depth drops to over 800 m. B. Water temperature fluctuation in autumn for shallow (5 m) and deep (45 m) sample areas (1972). This general trend occurred in 1969, 1971, 1973, and 1974; apparently characteristic of the season and region. Note the shallow area is more susceptible to short-term temperature fluctuations.

water visibility was often poor (less than 5 m) due to surge-induced turbidity.

The deep fringe site near Bimini selected as typical of the edge of the Bahama Bank was the Kinks area, site 3. It lay 1 km due west of North Bimini, approximately 5 ha in area, and 17-20 m in depth. The habitat was characterized by Gulf Stream water (visibility often exceeding 30 m), a sand bottom dotted by large coral and/or rock formations (1-2 m high), with sea whips, sea fans, and sponges interspaced. The dens included large, deep rock or coral structures, and often 10 or more lobsters were found together. Lobsters were numerous throughout the sampling period.

The migratory build-up site selected for sampling was one perennially associated with the appearance of autumnal migrants, site 13. It was near the west shore of South Bimini, 6-8 m in depth. Low rock ledges and sparse coral provided habitat approximately 2 ha in area isolated by 1-3 km of open sand from other habitable sites. Water quality varied from Gulf Stream clear (> 30 m) to very turbid (< 3 m), depending on wave surge and tidal conditions. (Sample area habitats indicated in Fig. 2A.)

The shallow areas (banks, migratory build-up) were more vulnerable to temperature fluctuations (implicated in the mass migration, Herrnkind et al., 1973) due both to loss of heat to the atmosphere during cold spells and the reduced effect of the Gulf Stream along the fringe of the western bank edge (Fig. 2B).

Habitat type was loosely correlated with depth; habitat description given for the bank sampling area and Kinks area are typical for shallow and deep areas in general, although exceptions were found.

All of these sample areas were relatively free from the heavy fishing pressure seen in areas further from Bimini. No trap fishing occurred at the main collection sites, 3, 13, and 11 nor at 1, 2, 12, 4, and adjacent regions. Occasional bully-netting occurred in the depths less than 5 m adjacent to sites 4 and 12. Extensive bully-netting took place during diurnal mass movements in 1969 and 1971 along the migratory pathway just inshore of the 6-m depth contour shown in Figure 1. We observed no lobstering by skin or SCUBA divers at any of the primary sites. As a result, we feel our samples were not influenced directly by fishing effort, although they certainly were subject to long term influence from cropping of large size classes and total population decline caused by heavy trap fishing in the (then) international waters adjacent to Bimini.

METHODS

We captured lobsters by tail snare, consisting of a hollow stainless steel tube, slightly curved at the far end, through which a wire cable was run and tied back forming a noose about 15-20 cm in diameter. Encircling the lobster's abdomen with the noose and drawing up tight on the cable at the diver's end firmly held the lobster without contacting the pereiopods or antennae, ensuring capture with minimal injury to body parts. The 1.5-m snare allowed probing of deep ledges and dens for recluse lobsters with minimal effort and diver trepidation. The animals were placed in net bags and brought to the surface for examination and tagging. Usually, every animal in a den was removed and examined. Infrequently it was necessary to hand capture a particularly seclusive or agitated lobster.

Morphometric measurements, including the carapace length (distance between the supra-orbital spines and the posterior edge of the cephalothorax) to the nearest mm, total weight (live weight to the nearest 10 g on a spring scale) and abdomen length (distance from the anterior edge of the dorsal surface of the abdomen to the posterior edge of the telson, abdomen flat, to the nearest 1 mm) were taken.

External carapace fouling was measured subjectively and values of clean (no macroscopic fouling), slightly fouled (fouled spots), fouled (scattered encrustations), and heavily fouled (encrustation solid over large areas) were assigned. External fouling included calcareous deposits, algae, bryozoans, foraminiferans, and barnacles. Presence of fouling was used as a rough indicator of time since last molt for the spawning cycle estimation.

Sexual condition was determined for each female by the presence of eggs and spermatophore masses; the latter classified as whole, partially eroded or completely eroded depending upon the ratio of smooth-rough spermatophore surface.

Throughout the study, lobsters were returned to the collection site after tagging; we did not examine ovarian state. For our purposes, a female in "reproductive condition" was one that carried eggs and/or a spermatophore of any type. The smallest gravid female we collected in the Bahamas was 74 mm in carapace length (weight approximately 400 g). Since size at sexual maturity seems to be highly variable depending upon geographic location (Smith, 1951; Sutcliffe, 1953), we chose this size as the lower limit defining a sexually "mature" female.

Approximately 2700 lobsters were collected. The MARS VI (*M*ulti-*A*ccess *R*etrieval *S*ystem) CDC 6400 computer at the Florida State University facilitated access to the data bank (Kanciruk, 1974).

RESULTS

Bank and Fringe Characterization

Both the shallow Bahama Bank surrounding Bimini (sites 5, 9, 10, 11, 12; Fig. 1) and the deeper fringing areas (sites 1, 2, 3, 4, 6, 7, 8; Fig. 1) were sampled between August and November, 1969 and 1971-74 to characterize the lobster populations found in these two diverse ecological habitats before, during, and after the fall migratory period (late October-early November). The bank habitat is typified by shallow (3-8 m sample depths) sandy bottom, intermittently covered by patches of sea whips, sponges, and *Thallassia* providing limited cover. The deeper fringe areas (10-25 m sample depths) surrounding the bank and adjacent to the sharp dropoff (30-49 m) have sandy or hard oolite substrates, with numerous rock ledges and/or coral patches (providing good shelter for resident lobsters) interspaced by large areas of open sandy bottom. Some of the major population characteristics of these two areas are listed in Table 2.

Lobsters inhabiting the bank during autumn are small, sexually immature, or sexually inactive. Only slightly more than half (57.8%) of all bank females examined were of a mature size, and none of the individuals carried eggs or any type of spermatophore (Fig. 3). Males were significantly larger than females (t' used due to unequal variances, male $\bar{x} = 79.4$ mm, female $\bar{x} = 75.3$ mm, $t' = 2.52$, df $= 232$, $p < .025$) and had a significantly larger variance in their sizes ($F = 2.24$, df $= 124, 108$, $p < .01$). The male-female ratio did not differ significantly from unity.

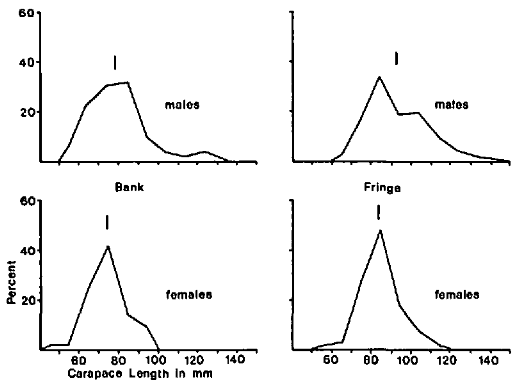


Figure 3. Size frequency curves for bank and fringe lobsters. Females in both areas are more tightly grouped about the mean, and are significantly (t' test used) smaller than males.

Lobsters inhabiting the deeper fringe areas are characteristically larger and more sexually active (Fig. 3); 91.0% of all females captured were sexually mature. Of these, 15.7% were gravid, and 42.4% showed signs of recent reproductive activity. The size frequency for males differed from the female size frequency in both mean (male $\bar{x} = 92.9$ mm, female $\bar{x} = 84.1$ mm, $t' = 13.1$, $p < .001$) and variance ($F = 2.64$, df $= 472, 587$, $p < .01$). During autumn, females were more numerous in the fringe areas, the sex ratio of 44.6% male/55.4% female significantly differs from unity at the $p < .005$ level (χ^2 goodness-of-fit). Bank males and females were significantly smaller than their counterparts from the Kinks area ($t' = 10.3, 23.8$, $p < .01$).

Reproduction and Depth

The average carapace length (CL) of male and female lobsters increased with depth. These data, along with reproductive information for all sample areas grouped, are presented in Table 3. Males were larger than females for each depth range, and both sexes increased in size as depth increased (Fig. 4). Over all depths, males averaged 88.2 mm CL while females averaged 81.4 mm. The frequency of mature females increased with depth, from 65.3% at 4 m or less to 94.8% at 20-25 m. The average size of females found to be in reproductive condition, however, remained relatively constant over all depths, averaging 87.4 mm CL.

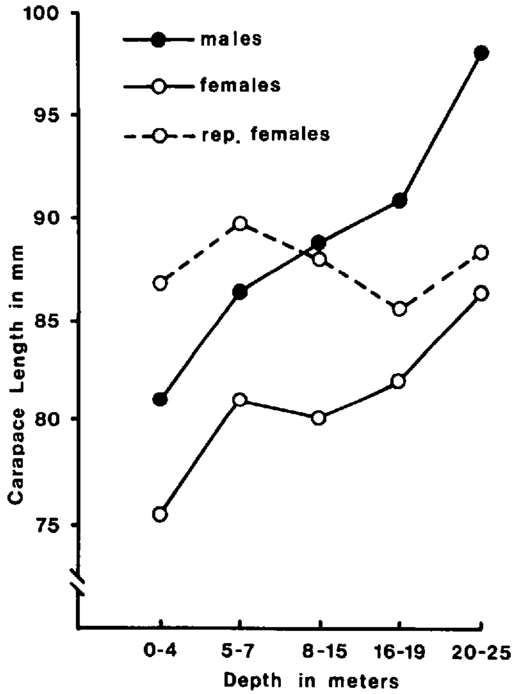


Figure 4. Relationship between lobster size and depth captured. Males on the average are larger than females for any given depth, and size increases with depth for both groups. Reproductive females sampled average approximately the same size irregardless of depth of capture. Note interrupted ordinate.

Sex ratios did not differ significantly from 1:1 in waters less than 16 m in depth. Females significantly outnumbered males in the deeper ranges (16-19 m, 20-25 m), 54.8% and 65.2% females, respectively, and over all depths (52.7%), Table 3.

Many facets of reproduction were correlated with depth. As noted above, the percentage of mature females increased with depth. In our shallowest sample, only 1% of mature females were in reproductive con-

Table 2. Bank and fringe characterization (CL = mean carapace length)

Area	N	% ♂	% ♀	CL ♂ (mm)	CL ♀ (mm)	% Eggs	% Eggs/sperm	% Mature	Avg. Depth (m)
Bank	235	53.6	46.4	79.4	75.3	0.0	0.0	57.8	4
Fringe	1065	44.6	55.4*	92.9	84.1	15.7	42.4	91.0	15

* Statistically significant difference in sex ratio (χ^2, $p \leq .05$).

dition, and none were found gravid. In 20-25 m, 60.0% of the mature females showed recent reproductive activity and 23.6% carried eggs (Table 3). As a measure of reproductive activity, the ratio of females with whole spermatophore or eggs to females with no eggs and eroded (partially or completely) spermatophore was used, and termed the "index of new reproductive activity," (INRA). In an area of new reproductive activity, there are proportionally fewer females with spent spermatophores and shed eggs, and therefore the index will yield a high value. In areas where new reproductive activity has declined, there will be relatively fewer females with whole spermatophores or eggs, more with spent spermatophores and shed eggs, and the index will be smaller. In our deepest samples (16-25 m) the index reached its highest values (0.87 and 0.70).

Reproduction and Size

Per cent of gravid females increased with female CL from 0% at ⩽ 70 mm to 15.6% at 91-100 mm (Table 4). The decrease to 9.5% in the ⩽ 101 mm class is not significant at the $\alpha = 0.05$ level due to the small sample size. Similarly, the per cent of females in reproductive condition increased from 0% at ⩽ 70 mm to 42.9% at ⩾ 101 mm CL.

Sex ratios varied with carapace length (Table 4B). The smallest group ⩽ 70 mm CL showed no significant deviation from the expected 1:1 ratio (χ^2). In the medium size classes (71-80 mm, 81-90 mm CL) females predominated significantly (about 40%♂, 60%♀). A reverse of this trend was exhibited in the larger sizes (91-100 mm, ⩾ 101 mm CL) where males were more abundant (63.7% and 79.7% males) and exceeded the maximum size of females.

Size-frequency curves for reproductive and non-reproductive females differ in that the former is shifted to the right, reproductives being larger than non-productives (non-reprod. $\bar{x} = 79.8$ mm, reprod. $\bar{x} = 87.1$ mm, $t = 12.35$, $df = 1438$, $p < .001$).

Table 3. Characterization by depth (CL = mean carapace length, Rep = reproductive, Mat = mature)

	Depth (m)					
	≤4	5–7†	8–15	16–19	20–25	All
N	165	1321	451	676	89	2702
% ♂	54.4%	48.4%	47.0%	45.1%	34.8%	47.3%
% ♀	45.5%	51.6%	53.9%	54.8%*	62.5%*	52.7%*
♂ CL	81.5 mm	86.8 mm	89.8 mm	91.1 mm	97.8 mm	88.2 mm
♀ CL	75.5 mm	81.3 mm	80.0 mm	82.8 mm	86.7 mm	81.4 mm
CL Rep ♀	87.0 mm	89.6 mm	88.0 mm	85.6 mm	87.6 mm	87.4 mm
% Mat ♀	65.3%	82.2%	76.2%	90.8%	94.8%	83.1%
% Eggs ‡	—	3.6%	11.0%	16.0%	23.6%	9.0%
% Eggs or sperm ‡	1.0%	14.6%	29.7%	37.1%	60.0%	24.9%
Index (INRA)	—	.49%	.64	.87	.70	.68

* Statistically significant difference in sex ratio (χ^2, $p \leq .05$).
† Primarily migratory build-up animals.
‡ Percentages based on females in mature size class only (≥ 74 mm CL).

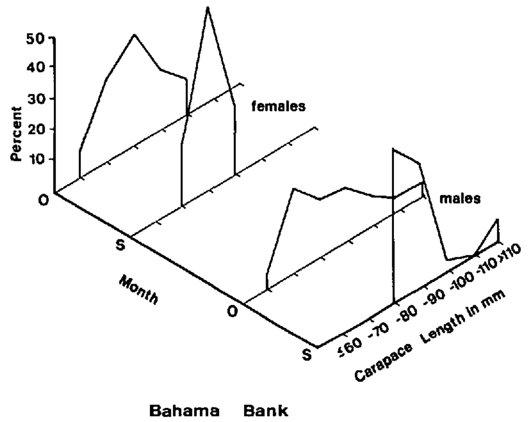


Figure 5. Bank animals, size frequencies for September (S) and October (O). Note an increase for both sexes in per cent of smaller lobsters and larger deviation around the mean for the October sample.

Autumnal Reproductive Sequence

Temporal reproduction and size frequency data for the banks, migratory build-up, and Kinks areas are presented in Table 5, and Figures 5, 6, and 7. The bank sample site adjacent to Bimini is characterized by the absence of reproductive activity, small size of individuals, and a 1:1 sex ratio, as noted for the general bank area. These parameters varied little over the autumn. Very few lobsters were found at the bank sampling areas during November, and the sample size is therefore small. The decrease in male size in November is not significant because of the small N.

In September, prior to the migratory influx, the sparse resident population of the migratory build-up area was typified by large average carapace size. The build-up animals found there in October and November were smaller and exhibited very low reproductive activity. These October and, especially, November collections reflect the characteristics of the mass migrants.

The Kinks area (typical of the fringe) was heavily sampled throughout the investigatory period. It was characterized by large lobsters, high sexual activity (16.7% to 64.3% of mature females in reproductive condition) and fluctuating sex ratios. Reproductive activity was highest during September (41.1% of mature females gravid), and lowest in November (6.7% of mature females gravid). Sex ratios for September and October were approximately 1:1, while in November, females were more abundant (56.4% of the sample, χ^2, $p < .05$). Overall, females were more abundant than males in the Kinks (55.3%, $p < .01$). During November, the INRA declined to 1.2 from a high of 2.6 in September.

Autumnal Spawning Cycle

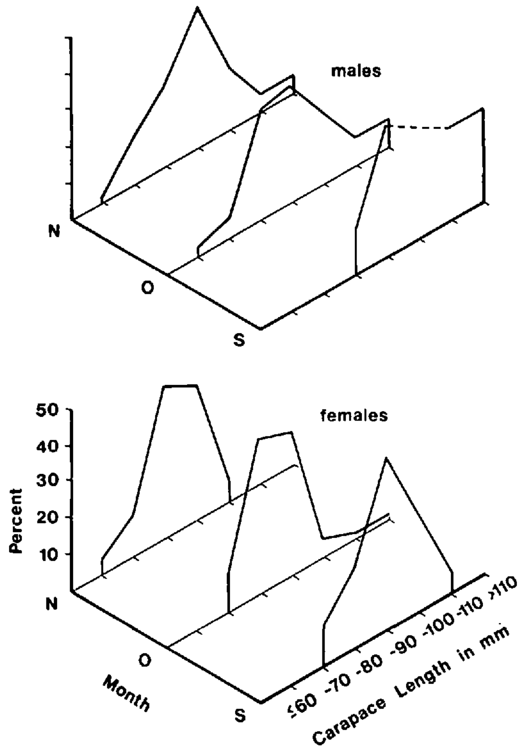
Comparison of the degree of fouling in different molt states indicated that as time since last molt increased, the per cent of lobsters showing external fouling also in-

Table 4. Characterization by size

	Carapace length (mm)					
	≤70	71–80	81–90	91–100	≥101	All
N ♀	154	521	570	159	42	1446
% Eggs	0.0	4.2%	9.8%	15.6%	9.5%	7.4%
% Eggs/sperm	0.0	12.1%	25.8%	44.4%	42.9%	20.7%

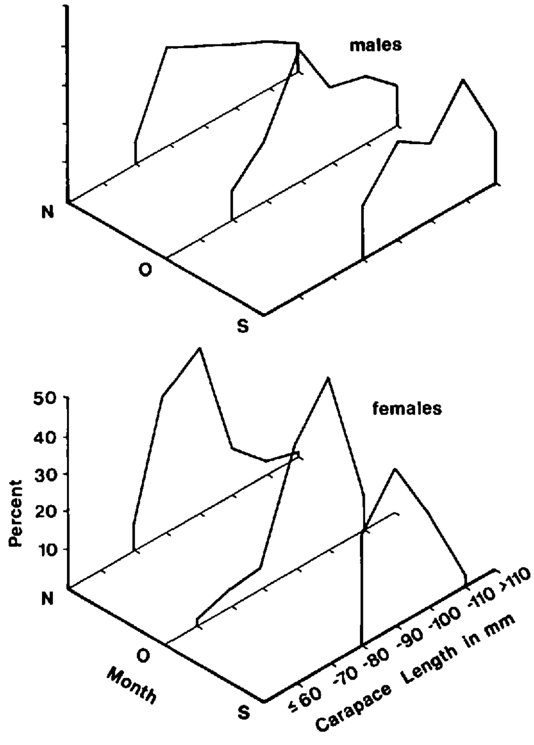
	Sex ratios by carapace length (mm)					
	≤70	71–80	81–90	91–100	≥101	All
N	286	805	989	440	207	2727
% ♂	46.2%	35.3%	42.4%	63.7%	79.7%	47.0%
% ♀	53.8%	64.7%*	57.6%*	36.3%*	20.3%*	53.0%*

* Statistically significant difference in sex ratio (χ^2, $p \leq .05$).



Migratory Buildup

Figure 6. Migratory build-up sampling site, size frequencies for September, October, and November. Note trend of a larger percentage of smaller lobsters represented in the later months, resulting from influx of mass migrants.



Kinks

Figure 7. Kinks sampling site, size frequencies for September, October, and November. Note same trends (to a lesser degree) as in Figures 5 and 6.

creased. It was thought that this rough indicator of time since last molt might be used to determine whether female lobsters at Bimini used one spermatophore to fertilize more than one batch of eggs (Sutcliffe, 1952, 1953). Reproductive females were found with either a whole spermatophore, partially eroded spermatophore (PE Sperm), or completely eroded spermatophore (CE Sperm). It has been suggested that a gravid female with a PE Sperm, after shedding her eggs, uses the remaining portion of her spermatophore to fertilize a second batch of eggs prior to her next molt.

The various states of reproductive condition with their associated fouling rates are presented in Figure 8A. If females fertilized two batches of eggs per spermatophore, this graph would represent a temporal sequence, with time running to the right on the abscissa. Females without eggs and with whole spermatophores are the most recently molted animals, and were infrequently fouled (6.3% of the sample). Females with eggs and PE Sperm showed a higher incidence of fouling (33.3%), and females without eggs and with PE Sperm were fouled 85.2% of the time. Females with eggs and CE Sperm, if they indeed represented a second laying with the same spermatophore, would be expected to show a similar or higher incidence of fouling than females without eggs and with PE Sperm.

Table 5. Characterization by time (CL = mean carapace length, Mat = mature, Rep = reproductive, INRA = Index of New Reproductive Activity)

Area	Month	N	%	%	CL (mm)	CL (mm)	% Mat	% Eggs	% Rep	INRA
Banks	Sept	57	52.6	47.4	83.3	76.1	55.6	—	—	—
	Oct	159	54.7	45.3	78.7	74.6	56.9	—	—	—
	Nov	19	47.4	52.6	73.3	78.6	70.0	—	—	—
	All	235	53.6	46.4	79.4	75.3	57.8	—	—	—
Migratory build-up										
	Sept	25	39.1	60.9	94.6	83.1	78.0	9.1	9.1†	—
	Oct	277	46.9	53.1	86.7	80.5	78.2	.9	12.2	.08
	Nov	444	49.5	51.5	85.0	78.4	76.8	.6	4.7	.14
	All	746	48.1	51.9	86.3	79.4	77.4	1.1	5.9	.11
Kinks	Sept	107	44.9	55.1	96.5	83.9	94.9	41.1	64.3	2.6
	Oct	240	45.8	54.2	91.8	82.3	89.2	20.7	48.3	2.5
	Nov	298	43.6	56.4*	88.8	82.5	89.3	6.7	16.7	1.2
	All	645	44.7	55.3*	91.2	82.7	90.2	17.7	36.3	2.0

* Statistically significant difference in sex ratio (χ^2, $p \leq .05$).
† Based on one gravid female.

Sampling indicated a fouling incidence of 42.6% for females with eggs and with CE Sperm, significantly lower than the 85.2% seen in females without eggs and with PE Sperm ($P < .005$, normal approximation to the binomial, corrected for continuity). No significant differences were observed between gravid females with PE Sperm and gravid females with CE Sperm (33.3% and 42.6% incidence of fouling, respectively), or between non-gravid females with PE Sperm and non-gravid females with CE Sperm (85.2% and 73.5% incidence, respectively) at the 0.05 level.

We felt that all reproductive stages, even recluse gravid females, were accurately represented in our diving collections. Therefore, frequency of occurrence of each stage was used as a gross indicator of time spent in each stage. Gravid females with CE Sperm were much more prevalent than gravid females with PE Sperm (21.4% to 9.4%, respectively). This would not be expected if some females with PE Sperm went on to use their spermatophores a second time and became berried females with CE Sperm; an equal or lower figure would be expected. We feel this is additional evidence that second fertilizations with the original spermatophore do not occur to any great extent in the autumn at Bimini. Therefore we regrouped the data, placing PE and CE Sperm into one category, and postulated a temporal sequence for spawning (Fig. 8B). If an egg carrying time of 25 ± 5 days is assumed (Sutcliffe, 1952), then the times for the other stages would fall as shown based on their sample representation: 6-9 days with whole spermatophore before egg laying, 20-30 days with eggs, 38-56 days without eggs and with eroded spermatophore. This yields a total of 84-125 days. However, the stage after the molt before the plastering of the spermatophore by the male is not included. If we assume an interval of 20 ± 5 days, this gives us a cycle length of 99-150 days or 3.7-2.4 molt cycles per year for 87.4 mm average CL females.

Reproductive Potential

Potential larval production by female size class was investigated. Egg bearing propensity (percentage of females in a given size class with eggs), egg carrying capacity (Mota-Alves and Bazerra, 1968), size frequency INRA, and contribution to total egg production by size class are given in Table 6. The size frequency of all females cap-

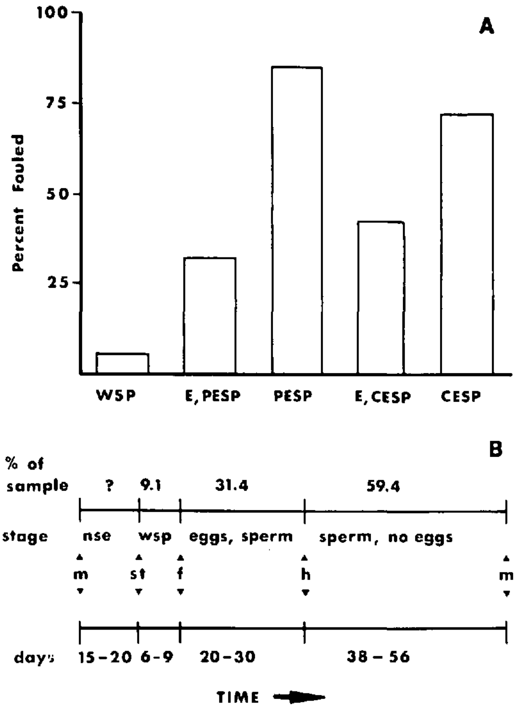


Figure 8. A. external fouling in relation to reproductive state. (WSP = Female with whole spermatophore; E, PESP = gravid with partially eroded spermatophore; PESP = non-gravid with partially eroded spermatophore; E, CESP = gravid with completely eroded spermatophore; CESP = non-gravid with completely eroded spermatophore.) Fouling is used as a rough estimate of time since the last molt. B. Per cent of sample in various reproductive states and estimated duration of each stage. (NSE = no spermatophore or eggs; WSP = whole spermatophore, no eggs, m = molt, st = spermatophore transfer, f = fertilization, h = hatching-release of eggs.)

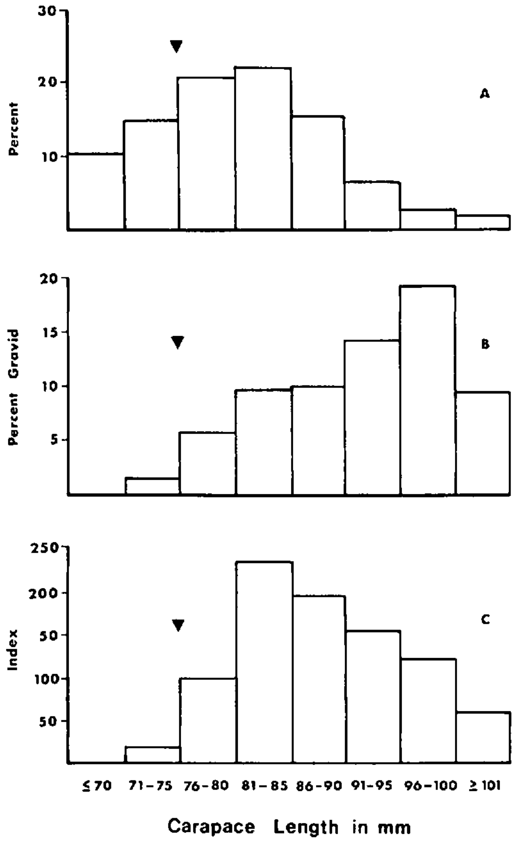


Figure 9. A. Size-frequency of all females. Arrow indicates minimum legal size. B. Per cent of all gravid females vs. size class. Note larger females were more frequently observed to carry eggs. C. Index of reproductive potential vs. carapace length. The bulk of egg production is generated by large females in the population even though they make up a small proportion of total females.

tured near Bimini is shown in Figure 9A. The propensity for females at Bimini in the fall to carry eggs, by size class, is seen in Figure 9B. Only 6% females in the 76-80 mm carapace length class were observed to carry eggs, while 19% of females in the 96-100 mm class were gravid; with larger females having a larger egg-carrying capacity (Table 6). To estimate the reproductive potential for each size class taking into consideration these variations between classes, an index of class reproductive potential was devised:

$$\text{Index} = (A \cdot B \cdot C)/D.$$

Where: A = # of females in class/total females

B = propensity of size class to carry eggs

C = egg carrying capacity of size class female

D = constant (31.27).

The constant, D, was chosen to set the 76-80 mm class's index to 100 as the stan-

Table 6. Index of reproductive potential (IRP)

	Size Class Of Female (CL in mm) $n = 1438$							
	≤ 70	71–75	76–80	81–85	86–90	91–95	96–100	≥ 101
A % of total	10.4	14.9	21.2	23.8	15.7	7.4	3.6	2.9
B % with eggs	0.0	1.9	5.9	9.6	9.7	14.0	19.2	9.5
C estimated # eggs carried when gravid ($\times 10^5$)	2.0	2.3	2.5	3.2	4.0	4.7	5.6	7.0
D IRP	0.0	20.5	100.0	233.8	194.8	155.7	123.8	61.7
E % of total egg production	0.0%	2.3%	11.2%	26.3%	21.9%	17.5%	13.9%	6.9%
F productivity (E/A)	0.0	.15	.52	1.1	1.4	2.4	3.9	2.4

dard. For example, the 76-80 mm class, which makes up 21.2% of all females captured, has a 5.9% propensity to carry eggs, and, when gravid, a female in this size range carries approximately 2.5×10^5 eggs. The calculated index is:

$$\text{Index} = [(.212 \times .059) \times (2.5 \times 10^5)]/31.27.$$
$$\text{Index} = 100.0.$$

Treated in this manner, our data provided a size class-index of reproductive potential relationship (Table 6, Fig. 9C). Females larger than 81 mm are more valuable in egg production than their numbers indicate. Females in the size class 76-80 mm CL represented 21.2% of all females collected, yet produced only 11.2% of the estimated total egg production for all females, a "productivity" rating (percentage of estimated total egg production of size class/size class percentage of all females) of $11.2/21.2 = 0.53$, (Table 6). The size class 96-100 mm represented only 3.6% of all females collected, yet contributed 13.9% of the estimated egg production (productivity $= 13.9/3.6 = 3.9$) making them approximately seven times as productive as a class. Newly mature, legally protected females (71-75 mm CL) made up 14.9% of all females collected, yet only produced an estimated 2.3% of the total egg production (productivity $= 0.15$) making them 26 times less productive than the 96-100 mm CL size class.

Discussion

Population Characterization

The shallow bank region near Bimini is populated by small, sexually inactive animals, at least during the fall. Such shallow areas have classically been thought of as juvenile areas abandoned when adult size is reached. We found that although bank animals are on the average smaller than fringe animals, young mature females were not infrequently found (57% of all females were $\geqslant 74$ mm), yet no reproductive females were observed. It seems that lack of autumnal reproduction on the bank is not just a function of small size. Females probably move to the deep bank fringe areas to mate, carry and release their eggs (Sutcliffe, 1952, 1953; Buesa Mas, 1965; Paiva 1968).

Our fringe sampling indicated sex ratios favoring females and relatively high rates of reproduction. In September, 39.3% of all females captured in the Kinks were gravid, a much higher rate than previously reported for September (Buesa Mass, 5%, 1965; Smith, approximately 5%, 1951). The Kinks area is not atypical in its high autumnal reproductive rate (females with eggs and/or sperm). Other fringe areas sampled also were so characterized: for the early autumn periods of 1971, 1972, 1973: 59% at Cat Cay (20 km south of Bimini), 39% at Turtle Rocks (5 km south of Bimini), 42% at Moselle Shoal (5 km north of Bimini), and 48% of females captured were in reproductve state at Great Issac Rock (35 km north of Bimini). It may be that the northwest Bahama Bank's high autumnal reproduction rate is unique or the other reports based mostly on trap sampling failed to discriminate local reproduction. For example, in one 1974 Kinks collection all but one of 22 gravid females occupied solitary dens, accounting for 39% of all solitary residences (54 dens). Non-gravid females and males frequently cohabit multiple dens; ratio 42.5%♂:57.5%♀ for 11 dens, $\bar{x} = 21.2$ lobsters per den. We suggest that trapping underestimates gravid females since they may be less likely to venture forth and/or enter a slat trap already holding lobsters. Possible advantages for the deeper fringe areas as mating and breeding sites are better shelter, more stable water conditions (temperature, salinity, reduced turbidity, and surge), and perhaps proximity to Gulf Stream current (larval transport).

Reproductive activity varied greatly for areas near Bimini (often only 1-2 km apart) and the disadvantages of lumping data collected from differing habitats are apparent. Lumping all our data and including mass migrants, a 9.1% reproduction emerges, much lower than the 39.3% rate for the Kinks. Results based on lumping of varied habitat types can be deceptive. Should management of the reproductive stock be deemed important in the future, we need knowledge of the important reproductive habitats and levels of annual reproductive activity for specific populations; e.g., Bimini area, Florida Keys, and other areas. Additionally, knowledge of the influx and efflux of reproductively active lobsters throughout each area will be necessary, thus local and area wide tagging-recapture programs will become essential.

Size and Reproductive Potential

Larger females were more frequently gravid than smaller females in our samples. This has been observed in other areas (Sutcliffe, 1952; Davis, 1974) and is not peculiar to the autumn or to Bimini. A distinction must be made between the smallest size at which a female is observed gravid, and the size of maximum reproductive potential. In Bimini, our smallest gravid female measured 74 mm in CL, yet maximum productivity was reached in the 96-100 mm CL size class. The legal size limit (3 in or 76 mm) seems to protect newly mature females, but perhaps does little to protect the reproductive potential of the population. As our index of reproductive potential indicates, these newly mature females (71-75 mm CL) generate only a small proportion (2.3%) of the total egg production estimated for the Bimini area in the fall.

In heavily fished populations, the size classes larger than 76 mm are drastically reduced (Davis, 1974; R. Warner, pers. comm.) with subsequent reduction in total egg production. Females are being cropped years before they reach their maximum reproductive potential. A closer look at management policies throughout the range of of this species may be in order.

Spawning Cycle

Data on incidence of fouling and frequency of occurrence of the various representative states suggest that autumnal reproduction does not involve a significant number of females who undergo multiple spawnings from one spermatophore. It seems that a female may erode all or only part of a spermatophore to fertilize her eggs independently of any second spawning. A possible objection to our frequency of occurrence data is that perhaps we are sampling the tail of the spawning cycle, and therefore are overestimating the occurrence of females with CE sperm. Our sampling was over a period of approximately 3 months, and the INRA was high during this period (Kinks average was 2.0, i.e., twice as many recently mated females as previously mated females), so that we feel that we were not merely sampling the tail-end of a cycle based on two spawnings per spermatophore cycle.

General Summary and Conclusions

Data over 5 years show that significant reproductive activity by *P. argus* occurs regionally along the edge of the northwest Great Bahama Bank well into autumn. The highest proportions of gravid females and the highest indices of recent mating occur in depths of 10 to 25 m on hard bottom habitat providing good shelter. Reproduction declines from September through November suggesting seasonal termination of mating and egg production at the onset of winter. We expect that reproduction here resumes and peaks in the spring much as in the Florida Keys, and the Bahamas generally (Smith, 1951), although we have no extensive data from the period December to July. Recent (1975) collections from the Kinks and migratory build-up area in mid-July show a pattern similar to early September suggesting continual reproduction through the summer.

The relationship of autumnal mass migrations to the reproductive population is not fully clarified, but our data are suggestive. First, the high September INRA is retained in the deep areas (e.g., Kinks) into late October after the "premigratory" build-up occurs in adjacent shallower regions. The migrants show very low or no reproductive activity and those moving into the deeper areas do not modify the INRA even though they inflate the population number and reduce the mean male and female average size. In November, the INRA shows a marked decline resulting from changes in relative numbers of new and eroded spermatophore bearing females. The cause for this is probably cyclic seasonal changes in reproductive physiology but may also result from behavioral interactions with new immigrants (e.g., mating interference, den crowding, induced emigration).

Comparison of Bimini spiny lobster reproduction with that of other regions is made difficult by the paucity of comparable data in the literature. Although similar techniques were used, data from the U.S. Virgin Islands cover only the period January to October (Herrnkind and Olsen, 1971; Olsen et al., in press). However, the inshore, hard substrate areas at 10 to 20 m depth show a high incidence of spermatophore bearing females in September (80+%) and October (60%). The low incidence (< 10%) of gravid females at this time suggests that females emigrate elsewhere to release larvae, probably offshore. The concomitant trend in sex ratio shows an increasing proportion of females in the fall inshore population as is true of the Kinks area. Generally, the U.S. Virgin Islands area, like Bimini, shows signs of autumnal reproduction. However, the two regions are ecologically quite different in depth distribution of habitat, seasonal character of water conditions, and lobster size frequency.

On going *in situ* research at Dry Tortugas promises comparative information from a more similar region (Davis, 1974). Preliminary data suggest peak reproduction in the spring with virtually none in autumn, even from habitats and depths comparable to Bimini (Davis, 1974, pers. comm.). This suggests significant differences in population character over the range of the species, requiring caution in applying trends from any particular area.

Finally, we advocate integration of *in situ* study of sample areas within broader regions examined through standard trap techniques. It is only in this way that the behavioral and ecological mechanisms underlying population dynamics can be determined and used to provide comprehensive understanding of the species.

Acknowledgments

We recognize that the information presented here could not have been obtained without the extraordinary efforts of the Spiny Lobster Undersea Research Project team (SLURP) particularly J. Halusky and R. McLean, and including R. Bill, A. Rudloe, P. Hamilton, M. Sinclair, R. McWilliams, G. Stanton, M. Halusky, C. Combs, R. Frehsee, and F. Walton. We owe much to Capt. J. Barber and Engineer R. Millender of R/V TURSIOPS. The directors of the Lerner Marine Lab during 1969-74, R. Mathewson and J. Tyler, and their staff provided fine field facilities. We thank W. Garcia, D. Schroder, S. Scoville, and W. Hamner and his research group for assistance at Bimini. Research support was provided by the National Science Foundation through grant GA-30813 to William Herrnkind and funding for R/V TURSIOPS, Florida State University; by financial assistance from the Department of Natural Resources, State of Florida, to Paul Kanciruk; and by a National Geographic Society grant to William Herrnkind. This is contribution No. 67 from the Tallahassee, Sopchoppy and Gulf Coast Marine Biological Association.

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